Project goal

Teach a computer to finish a clean lap.

Then give it a different road.

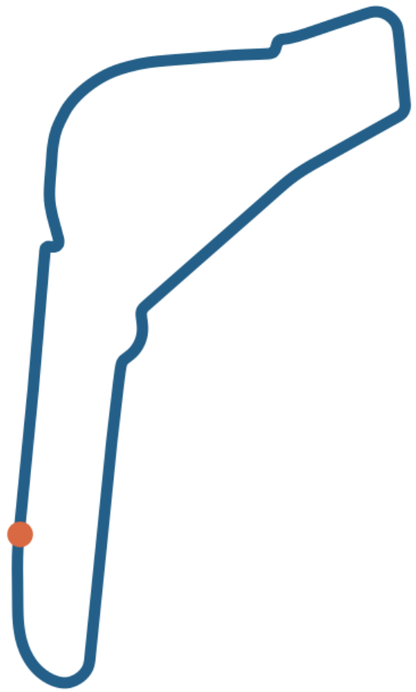
A completed lap with zero road departures comes before a faster time.



Actual simulator replay at Monza. Optional clip: 04_monza_full_lap.gif.

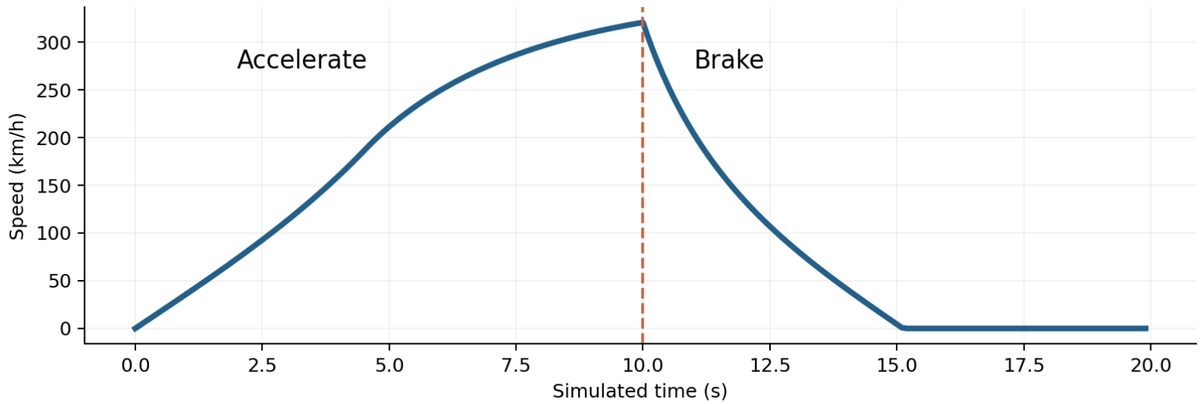
Track geometry and vehicle physics

THE ROAD



Shape, bends and boundaries

THE CAR



Steering and pedals change position, direction and speed.

Road map: bundled Monza geometry. Physics plot: controlled 10 s acceleration, then 0.6 braking, 0.1 s steps; not a racing benchmark.

Simulator



Road limits

Is the car on the road?

Lap progress

How far has it travelled?

Time

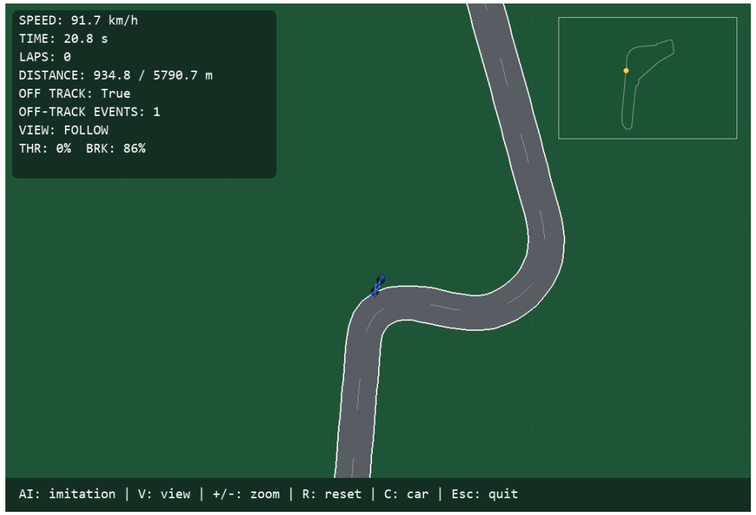
Has it completed a lap?

Track + shared physics + session state. Actual follow-camera screenshot; map is not a real onboard image.

Three driving systems

01

Naive learned pedals



Random Forest pedals

Rules still steer and handle launch.

91.8 s

13 departures

GIF 01_no_speed_planner.gif

02

Assisted learned pedals



Same Random Forest

Add a rule-based speed planner.

205.4 s

0 departures

GIF 02_with_speed_planner.gif

03

Learned full control



Neural steering + ExtraTrees pedals

New simulated-teacher examples.

209.1 s

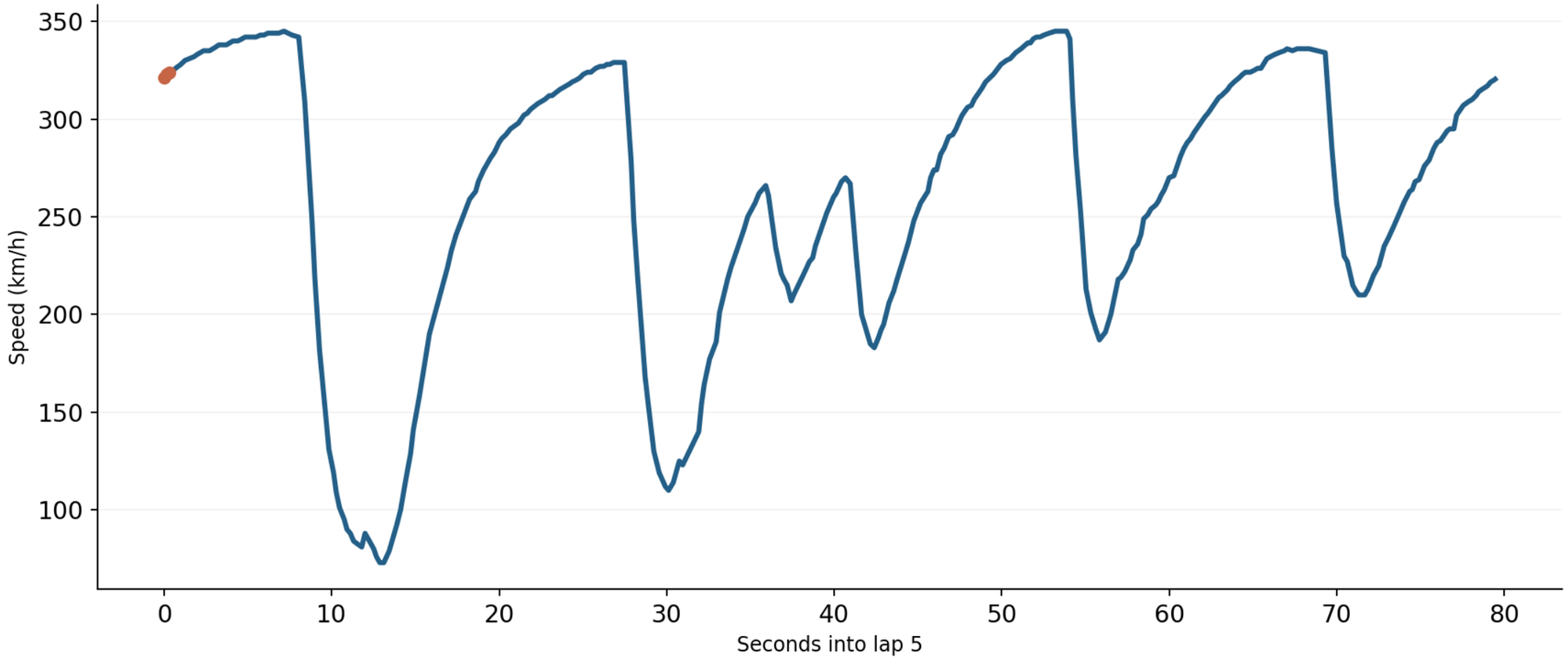
0 departures

GIF 03_learned_full_control.gif

GIFs 01-03: independent replays near the same Monza turn, not a synchronized race. Times use 0.1-second steps.

Recorded driving examples

One row records one moment of a lap.



590 moments

in lap 5 alone

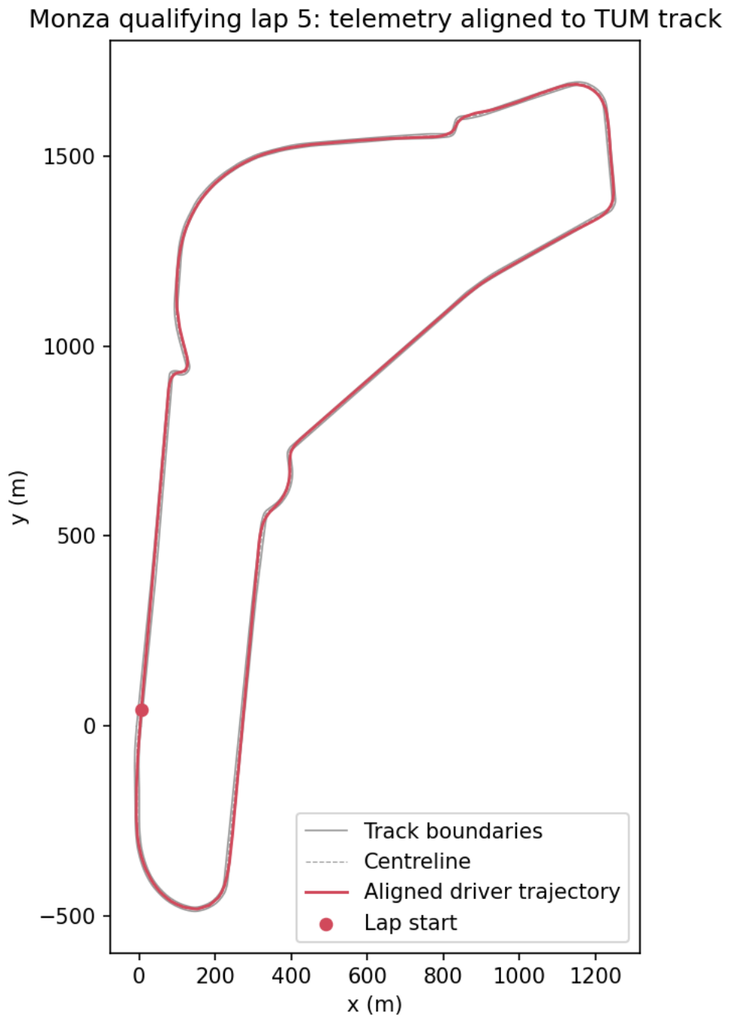
1,774 moments

across laps 5, 11 and 14

At 0.154 s	323.0 km/h	100% throttle	Brake off

Actual saved lap 5. Public channels were synchronized/interpolated; saved moments are not independent trials.

Coordinate alignment



Same moment. A useful road reference.

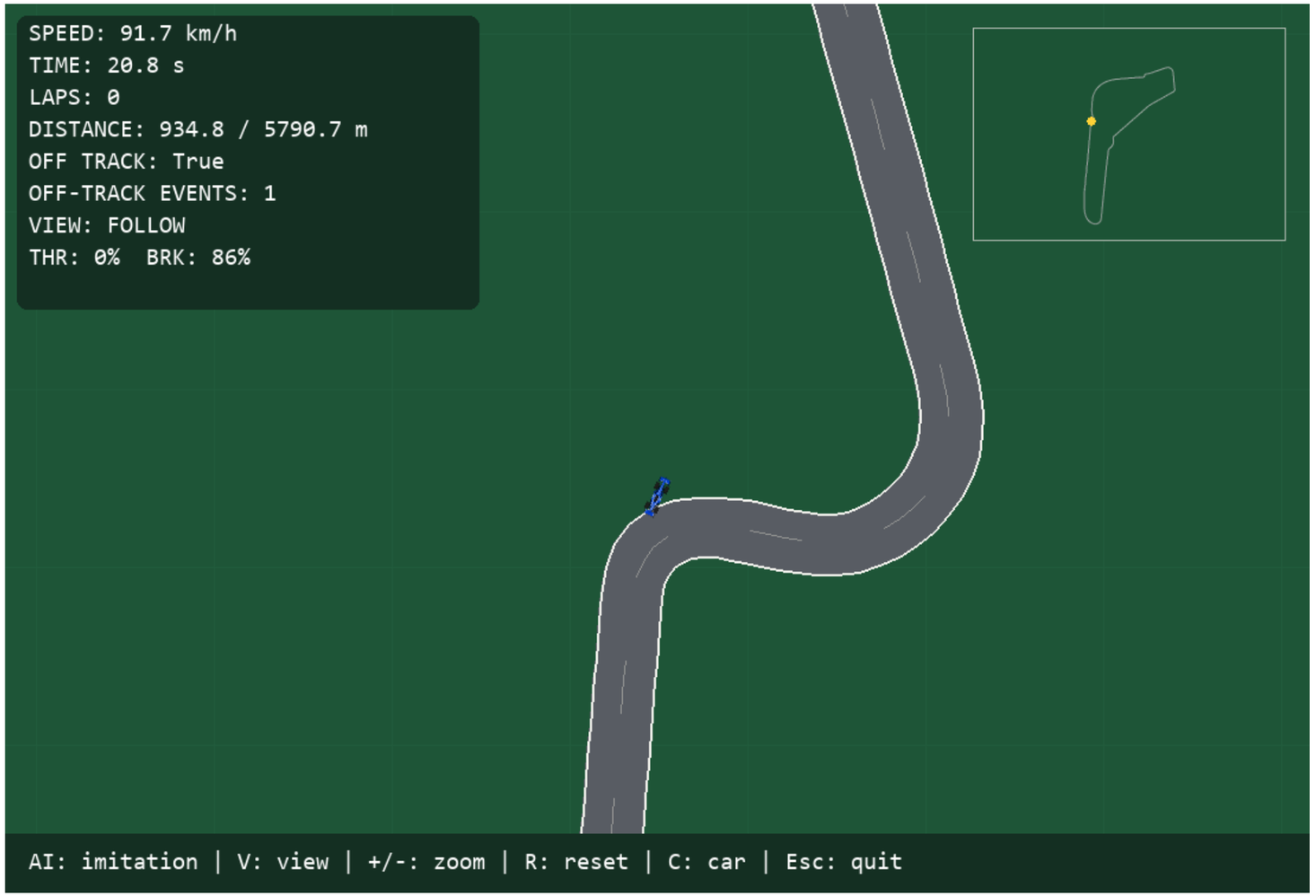
Before: position in the recording's coordinates.

After: 49.1 m along our road, 2.26 m right of its centre.

Keep speed and recorded pedals attached to that road position.

Actual lap-5 alignment output. A full training trajectory estimates the transform; a single point cannot determine it.

What if the model chose the pedals?



Random Forest

Learns throttle and brake from matched examples.

Rules still handle steering, launch and pedal cleanup.

91.8 s

Monza completion

13 road departures

Stage 1: original Random Forest, 500 m preview. GIF: 01_no_speed_planner.gif. Invalid completion; not a time improvement.

What if we added speed assistance?

205.4 s

Monza: valid lap

0 departures

Keep the model.
Add a speed planner before the corners.

Rules still steer.



Stage 2: same original learned-pedal model. GIF: 02_with_speed_planner.gif. The speed planner can override its proposals.

What if the models chose all the controls?

40,000 simulated situations

Starts, corners and recovery, labelled by a rule-based teacher.

Neural network

Chooses steering

ExtraTrees

Chooses throttle or brake

209.1 s / 0 departures

No runtime driving rules; actuator limits remain.



Stage 3: expert imitation, not reinforcement learning. GIF: 03_learned_full_control.gif. Teacher labels are used during training only.

Model checks and tuning

Train on one whole lap.
Check on the other. Swap.

18 settings across three model families.
A fixed 50/50 combined predictor.
29 existing controller tests passed.

Random Forest



0.0538 +/- 0.0011

50/50 combined model



0.0545 +/- 0.0003

Support-vector model



0.0600 +/- 0.0021

Neural network



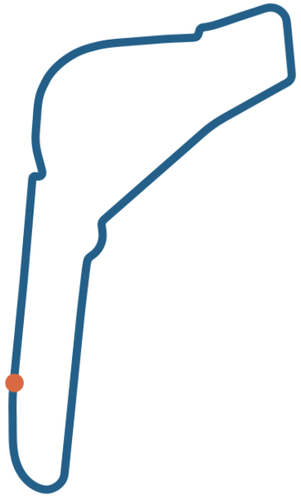
0.0975 +/- 0.0009

Random Forest selected before Spa testing.

Pedal prediction MAE on a 0-1 scale; lower is better. Mean +/- sample SD over two whole-lap selection folds, reused for tuning; not final-test accuracy.

New circuits and the final test

Monza, Italy



TRAINING ROAD

209.1 s

Examples come from this circuit.

Silverstone, UK

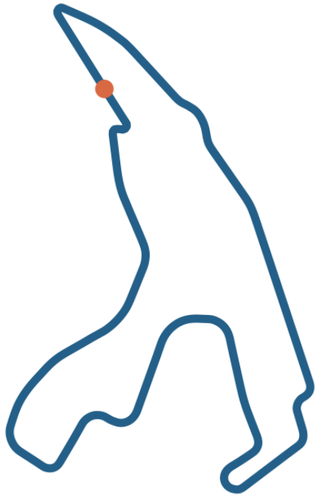


FIRST TRANSFER CHECK

266.3 s

Excluded from model training.

Spa-Francorchamps



FRESH FINAL TEST

294.3 s

Belgium. Frozen models; no test tuning.

All shown stage-3 laps completed with zero departures. Circuit maps use bundled geometry; Spa identity: formula1.com/en/racing/2026/belgium.

Benchmarks and performance

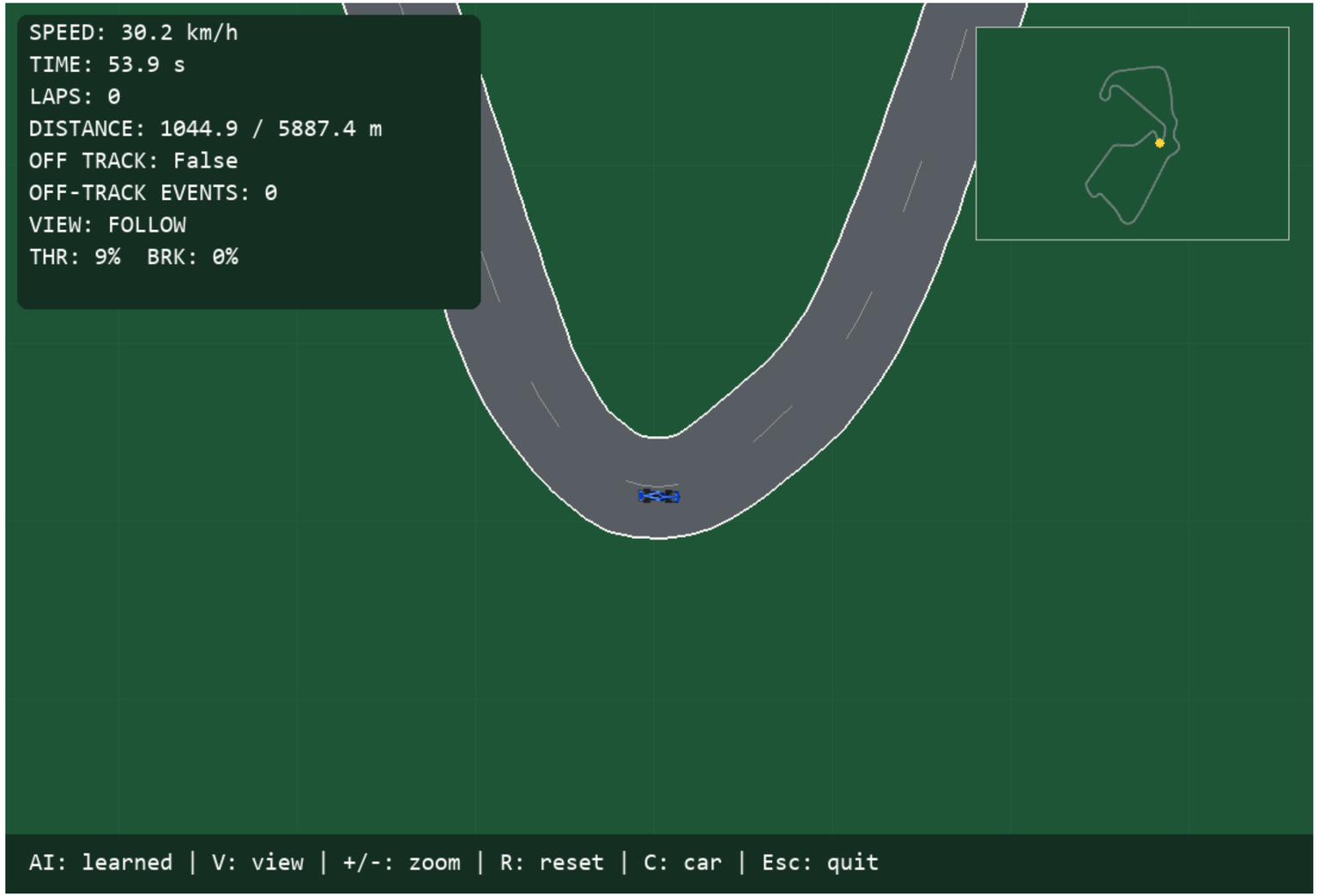
Same start, physics and time step. Eight complete driving configurations; all three circuits.

Driving configuration	Steering	Pedals	Speed planner	Monza	Silverstone	Spa
Rules only	Rule	Rule	Yes	**208.7 s** Clean lap	**258.3 s** Clean lap	**287.4 s** Clean lap
Original RF, no speed planner	Rule	Original RF	No	**91.8 s** 13 departures; invalid	**Timeout (600 s)** 5 departures; invalid	**Timeout (600 s)** 9 departures; invalid
Original RF + speed planner	Rule	Original RF	Yes	**205.4 s** Clean lap	**256.5 s** Clean lap	**284.6 s** Clean lap
Learned steering + pedals	Neural network	ExtraTrees	No	**209.1 s** Clean lap	**266.3 s** Clean lap	**294.3 s** Clean lap
Tuned RF + speed planner	Rule	Tuned RF	Yes	**205.5 s** Clean lap	**253.7 s** Clean lap	**282.0 s** Clean lap
Tuned SVR + speed planner	Rule	Tuned SVR	Yes	**204.1 s** Clean lap	**253.6 s** Clean lap	**281.9 s** Clean lap
Tuned neural pedals + planner	Rule	Tuned neural net	Yes	**199.1 s** Clean lap	**248.1 s** Clean lap	**276.8 s** Clean lap
50/50 RF + SVR + planner	Rule	RF + SVR blend	Yes	**205.1 s** Clean lap	**254.7 s** Clean lap	**281.2 s** Clean lap

Clean = completed, zero departures. Other completions are invalid; timeout = no completed lap. Only compare speed for clean laps.

All attempts: output/all_configurations. Pedal-prediction error is a separate metric. Full-control imitation also changes training and steering.

What we built



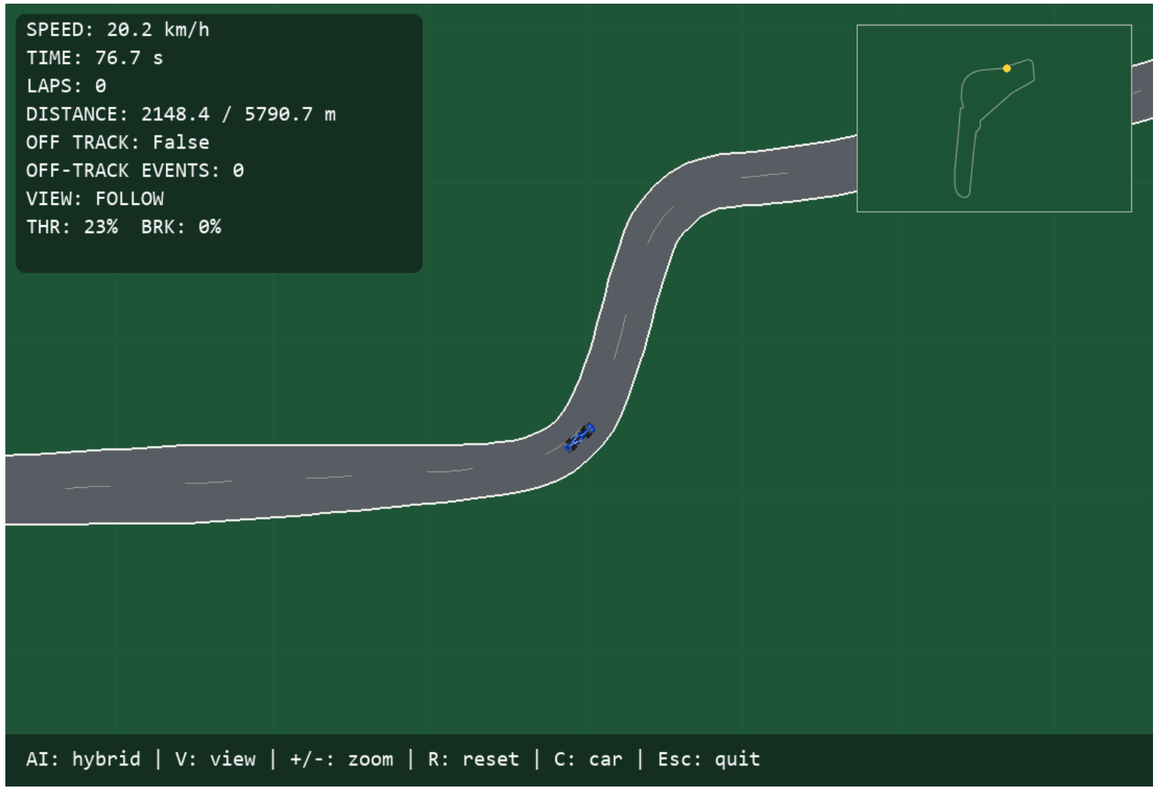
Seven components. One working simulator.

Recorded examples became learned controls.

The full learned driver completed clean laps on new circuits.

Approximate physics and track limits; a simulated teacher; one deterministic final-test start. Real-car performance and broad reliability remain unestablished.

Monza: additional turn screenshots



Assisted driver: second Monza turn



Learned steering and pedals at Monza

Actual simulator replays. Choose one as an alternative photo; these independent runs are not synchronized.

Silverstone: additional screenshots



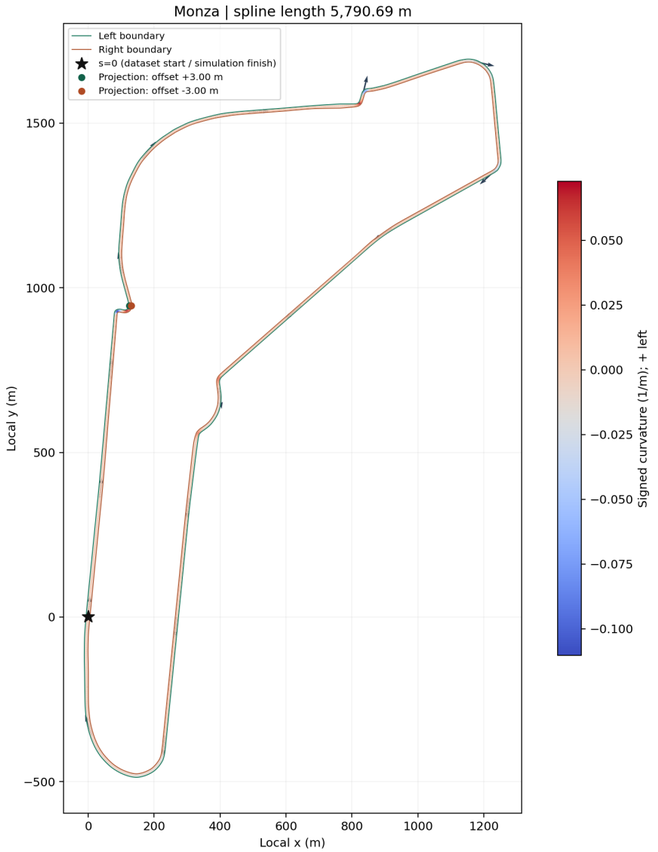
Assisted driver at Silverstone



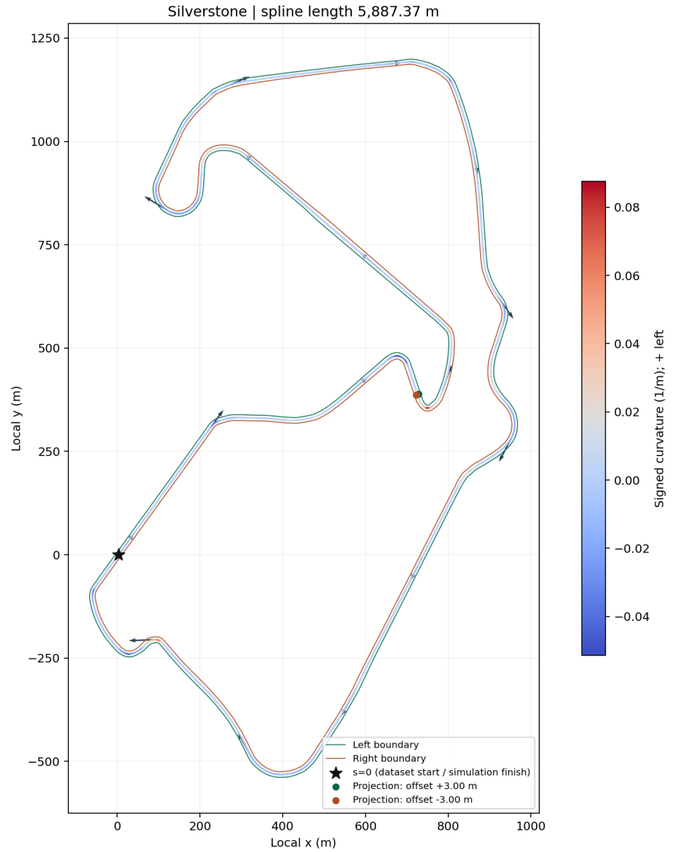
Learned full-control driver at Silverstone

Silverstone is the UK transfer circuit. These images are not Spa footage.

Track geometry: detailed reference plots



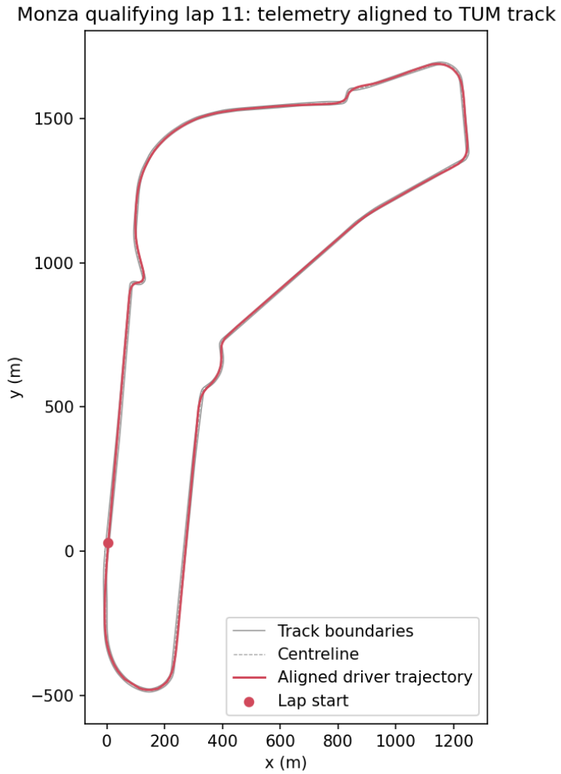
Monza: centreline, boundaries and curvature



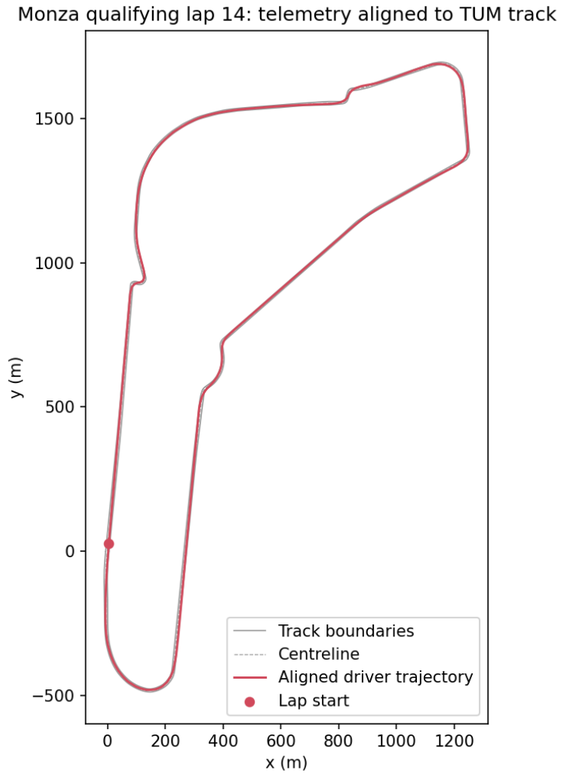
Silverstone: centreline, boundaries and curvature

Detailed project plots for an optional technical slide; use the simple maps in the main story for the audience introduction.

Alignment: additional recorded laps



Lap 11: transformed telemetry on the map



Lap 14: transformed telemetry on the map

The transform was fitted using training lap 5. Lap 14 was excluded from model fitting, then inspected during development; its scores are exploratory.

Full-lap animation references



Optional_Stage3_Monza_FullLap_10x.gif



Optional_Stage3_Silverstone_FullLap_10x.gif

PDF uses stills. Insert the real replay GIF files in Canva for motion; full-lap playback is nominal 10x speed.